Supplementary Material: Directional Prompting for Gaze-Referential Inference in Large Multimodal Models

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This document provides more details excluded from the main paper due to space constraints: construction details for CG-GRI, how the ground-truth arrows are defined, the arrow rendering style, the evaluated model identifiers, the statistical testing protocol, the LMM-generated-arrow prompt, and an analysis of `Gemini-3.1-pro`'s sensitivity to UniGaze-predicted gaze arrows.

1 Construction of CG-GRI

CG-GRI (N=483) is derived from VideoAttentionTarget [1] (N=200) and GazeFollow [2] (N=283). Neither source provides GRI-ready candidate lists, so we construct the multiple-choice options and targets as follows. We refer to the two subsets as VAT-GRI and GF-GRI; they are reporting slices of one CG-GRI track rather than separate evaluation tracks.

VideoAttentionTarget (VAT-GRI). As consecutive frames within a clip are visually similar, we sample by gaze sequence or clip and retain at most one frame per clip. For each selected frame, automatic screening uses YOLOv10L [3]: the observer head box must be valid; detections overlapping the observer box are removed; the annotated gaze point must fall within at least one remaining box; and at least two candidate boxes must remain. Candidate descriptions are drafted by an LMM and human-edited so that instances of the same class remain visually distinguishable. Every retained instance is human-annotated for observer identity, candidate list, and the target indicated by the original VAT gaze point; ambiguous images are discarded. Ground-truth arrow origins are also human-annotated eye points, while endpoints remain the original VAT gaze points. This process yields 200 VAT-GRI instances.

GazeFollow (GF-GRI). GazeFollow provides web images with gaze-region annotations but no gazed-object labels. We randomly sample 1,000 images and retain those in which the observer's eyes and gaze direction can be judged. Images with occluded or unclear eyes are discarded. For every retained image, candidate objects are drafted using YOLO and human-edited, and the gazed target is human-annotated using the original GazeFollow heatmap. Ground-truth arrows use the original eye annotation as the origin and the original gaze point as the endpoint. This process yields 283 GF-GRI instances.

Shared protocol. The target observer is marked by a red unfilled bounding-box outline in every CG-GRI condition, including the RGB baseline. The box disambiguates the observer and selects whose gaze is predicted, but it is not used as the arrow origin. Among detected faces overlapping the box, we retain the prediction whose projected origin is closest to the reference head. The question lists the human-annotated candidate descriptions and asks which candidate the marked observer is looking at.

2 Ground-Truth Arrows

The ground-truth arrows ($\mathsf{A(GT)}$ and $\mathsf{A_{txt}(GT)}$) are drawn from an origin to an endpoint, defined per track as follows.

FG-GRI. The ground-truth arrows are human-annotated. The endpoint is the annotated gaze point of each instance, collected with Label Studio. The origin is the midpoint between the observer's eyes produced by the UniGaze pipeline and verified by annotators.

CG-GRI. Both ends come from the source annotations: the origin is the human-annotated eye point (VAT-GRI) or the original GazeFollow eye annotation (GF-GRI), and the endpoint is the original gaze point of each dataset (Section 1).

Since the endpoint is the annotated gaze point, a ground-truth arrow points to the target by construction. Accuracy with $\mathsf{A(GT)}$ therefore measures how well an LMM reads an arrow that points at the target, and should be interpreted as an oracle-cue reference rather than as an upper bound for a perfect gaze estimator.

3 Arrow Rendering

Visual arrows are drawn with OpenCV's anti-aliased arrow primitive in pure red (RGB $(255, 0, 0)$) over a thin white outline, so that the arrow stays visible on dark and bright backgrounds. On FG-GRI, whose images are 448×448 pixels, the shaft is 1 pixel wide for UniGaze arrows and 2 pixels wide for ground-truth arrows, and the arrowhead is a fixed fraction of the arrow length. The line and point controls ($\mathsf{RGB{+}L}$, $\mathsf{RGB{+}P}$) use the same red-and-white style, without the arrowhead and with a filled disk, respectively. The depth-colored arrow ($\mathsf{D{+}A}$) is drawn on the depth image with the color of each segment given by the interpolated ray depth.

4 Evaluated LMMs

Table 1 lists the model identifiers used in all experiments. The proprietary LMMs and `Gemma-4-26B` were accessed through commercial APIs; all LMMs receive the same zero-shot multiple-choice prompt, and the JSON `answer` field of each response is scored by exact match against the ground-truth label.

Table 1. Model names in the main paper and the identifiers used.

Name in paper	Identifier
Qwen3-VL-8B-Instruct	qwen3-vl-8b-instruct
Qwen3.5-27B	qwen3.5-27b
Llama-3.2-11B-Vision	llama-3.2-11b-vision-instruct
Gemma-4-26B	gemma-4-26b-a4b-it
GPT-4.1	gpt-4.1
GPT-5.1	gpt-5.1
GPT-5.2	gpt-5.2
Gemini-2.5-flash	gemini-2.5-flash
Gemini-3.1-pro	gemini-3.1-pro-preview

Table 2. RGB+A(UniGaze) vs. RGB: accuracy (%), paired difference (percentage points) with 95% bootstrap CI, McNemar p, and Holm-adjusted p. n is the number of paired instances.

Model	n	RGB	RGB+A	Difference [95% CI]	p	Holm p
FG-GRI						
Qwen3-VL-8B-Instruct	904	45.6	72.1	+26.5 [+22.9, +30.2]	<0.001	<0.001
Llama-3.2-11B-Vision	904	41.5	62.1	+20.6 [+17.1, +24.0]	<0.001	<0.001
Gemma-4-26B	904	50.8	79.6	+28.9 [+25.2, +32.4]	<0.001	<0.001
Qwen3.5-27B	904	44.6	72.2	+27.7 [+24.1, +31.2]	<0.001	<0.001
Gemini-2.5-flash	904	47.8	79.8	+32.0 [+28.3, +35.6]	<0.001	<0.001
GPT-4.1	904	45.5	79.4	+34.0 [+30.1, +37.7]	<0.001	<0.001
GPT-5.1	904	44.8	83.7	+38.9 [+35.1, +42.7]	<0.001	<0.001
GPT-5.2	904	41.0	78.1	+37.1 [+33.2, +40.9]	<0.001	<0.001
Gemini-3.1-pro	904	55.5	86.4	+30.9 [+27.5, +34.3]	<0.001	<0.001
CG-GRI						
Qwen3-VL-8B-Instruct	482	61.2	71.0	+9.8 [+6.2, +13.5]	<0.001	<0.001
Llama-3.2-11B-Vision	483	51.6	56.9	+5.4 [+1.7, +9.1]	0.008	0.046
Gemma-4-26B	483	72.3	74.9	+2.7 [-1.0, +6.4]	0.188	0.563
Qwen3.5-27B	480	73.1	77.7	+4.6 [+0.8, +8.3]	0.025	0.113
Gemini-2.5-flash	483	73.1	76.2	+3.1 [-1.2, +7.5]	0.192	0.563
GPT-4.1	483	68.1	74.1	+6.0 [+2.1, +10.1]	0.005	0.037
GPT-5.1	483	66.0	72.5	+6.4 [+2.3, +10.8]	0.004	0.035
GPT-5.2	483	71.4	76.2	+4.8 [+1.0, +8.7]	0.023	0.113
Gemini-3.1-pro	483	78.7	77.2	-1.4 [-5.8, +3.1]	0.582	0.582

5 Statistical Testing

For each dataset, we test RGB+A(UniGaze) against RGB for the nine LMMs with a paired two-sided McNemar test (continuity-corrected) and apply Holm correction over these nine tests. Table 2 reports the paired accuracy difference with a 95% bootstrap confidence interval (10,000 resamples), the raw p-value, and the Holm-adjusted p-value. On FG-GRI, all nine gains remain significant after correction. On CG-GRI, six gains are significant before correction and four after (`Qwen3.5-27B` and `GPT-5.2` are no longer significant). Other comparisons in the paper, including the stratified analyses, use uncorrected tests and should be read as exploratory.

6 LMM-Generated-Arrow Prompt

The first stage of RGB+LMM-gen receives no answer choices and returns a gaze ray that is rendered on the image before the plain multiple-choice prompt is applied.

Table 3. Model outcomes on the Arrow-May-Help and Arrow-May-Harm sets of CG-GRI.

Diagnostic subset and outcome	`Gemini-3.1-pro`	`Qwen3.5-27B`
Arrow-May-Help: RGB+A(UniGaze) correct (N=90)	**87/90 (96.7%)**	71/90 (78.9%)
Arrow-May-Harm: RGB+A(UniGaze) wrong (N=83)	**70/83 (84.3%)**	48/83 (57.8%)

```
Identify the target person's gaze direction in this image.
Do not name an object.
Return exactly one JSON object using a normalized 0-1000 image coordinate system:
{"start": [x, y], "end": [x, y]}.
The start must be near the target person's visible eyes, and the end must lie in the
gaze direction.
```

The JSON `answer` field of the second-stage multiple-choice response is scored by case-insensitive exact match against the ground-truth label.

7 Sensitivity of Gemini-3.1-pro to UniGaze-Predicted Gaze Arrows

The main paper reports a significant accuracy degradation for `Gemini-3.1-pro` in the 3-candidate stratum of CG-GRI, where RGB+A(UniGaze) underperforms RGB, while other LMMs such as `Qwen3.5-27B` see RGB+A(UniGaze) outperforming RGB. Motivated by this result, we analyze the effect of A(UniGaze) on `Gemini-3.1-pro` and `Qwen3.5-27B`.

We first consider two sets of samples: (a) the *Arrow-May-Help* set, covering samples for which at least one of the two LMMs predicts wrongly with RGB but correctly with RGB+A(UniGaze); and (b) the *Arrow-May-Harm* set, covering samples for which at least one of the two LMMs predicts correctly with RGB but incorrectly with RGB+A(UniGaze). Table 3 shows that these sets contain 90 and 83 samples, respectively.

For the Arrow-May-Help set, Gemini yields 16 more correct predictions than Qwen (87 vs. 71). This suggests that when the arrow helps, Gemini can more effectively benefit from it. The finding is quite different for the Arrow-May-Harm set. On this set, Gemini suffers 70 wrong predictions with RGB+A(UniGaze) out of the 83 samples that the arrow may harm, whereas Qwen suffers only 48 out of 83. In conclusion, Gemini is relatively more susceptible to arrow cues that may harm LMMs in CG-GRI tasks.

References

1. Chong, E., Wang, Y., Ruiz, N., Rehg, J.M.: Detecting attended visual targets in video. In: CVPR. pp. 5396–5406 (2020)
2. Recasens, A., Khosla, A., Vondrick, C., Torralba, A.: Where are they looking? In: NeurIPS. pp. 199–207 (2015)
3. Wang, A., Chen, H., Liu, L., Chen, K., Lin, Z., Han, J., Ding, G.: YOLOv10: Real-time end-to-end object detection. In: NeurIPS. vol. 37, pp. 107984–108011 (2024). https://doi.org/10.52202/079017-3429